

From Batch-Level Skew to Adaptive Service Zones

Rethinking fixed shuttle ownership in Project Silica

Question: How can a glass library adapt its service regions to a changing active workload without giving up spatial locality?

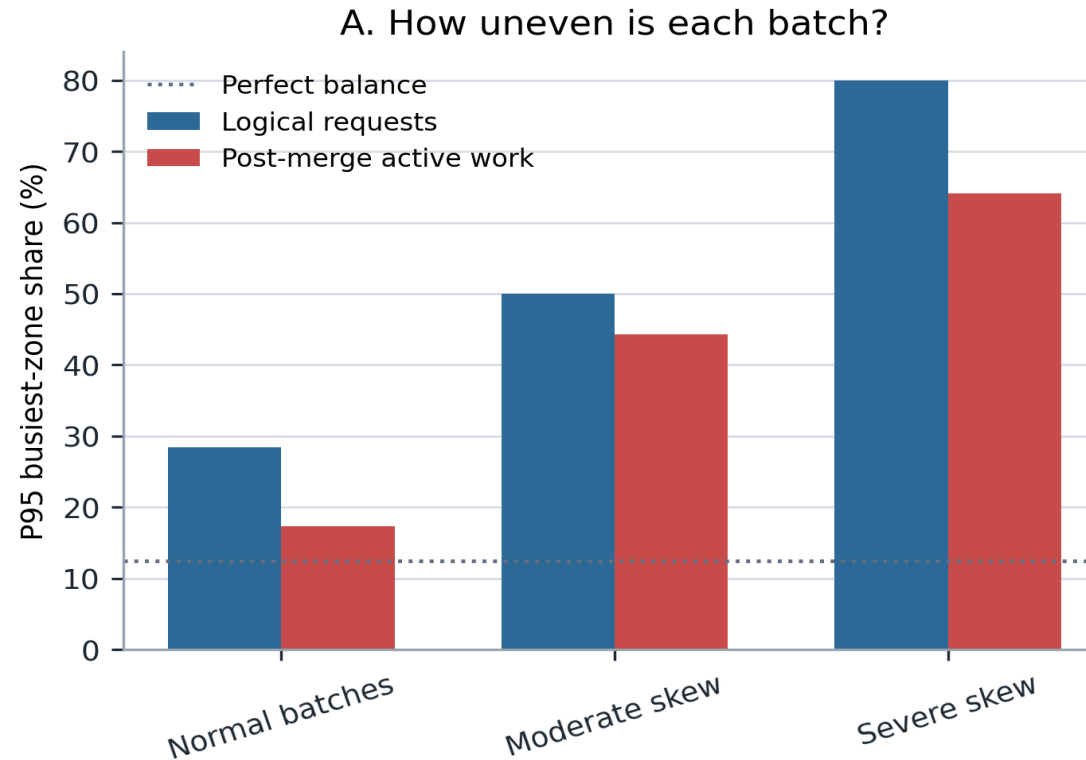
Presentation draft based on the current trace-derived and collision-free simulation results.

A Controlled View of Batch-Level Access Skew

- We replay the **same 100,000 trace requests** in 20 consecutive batches.
- Each batch contains **5,000 logical requests**.
- Only batch ordering changes:
 - Normal timestamp order
 - 50% target hot-zone share
 - 80% target hot-zone share
- Request merging is performed **within each batch**.

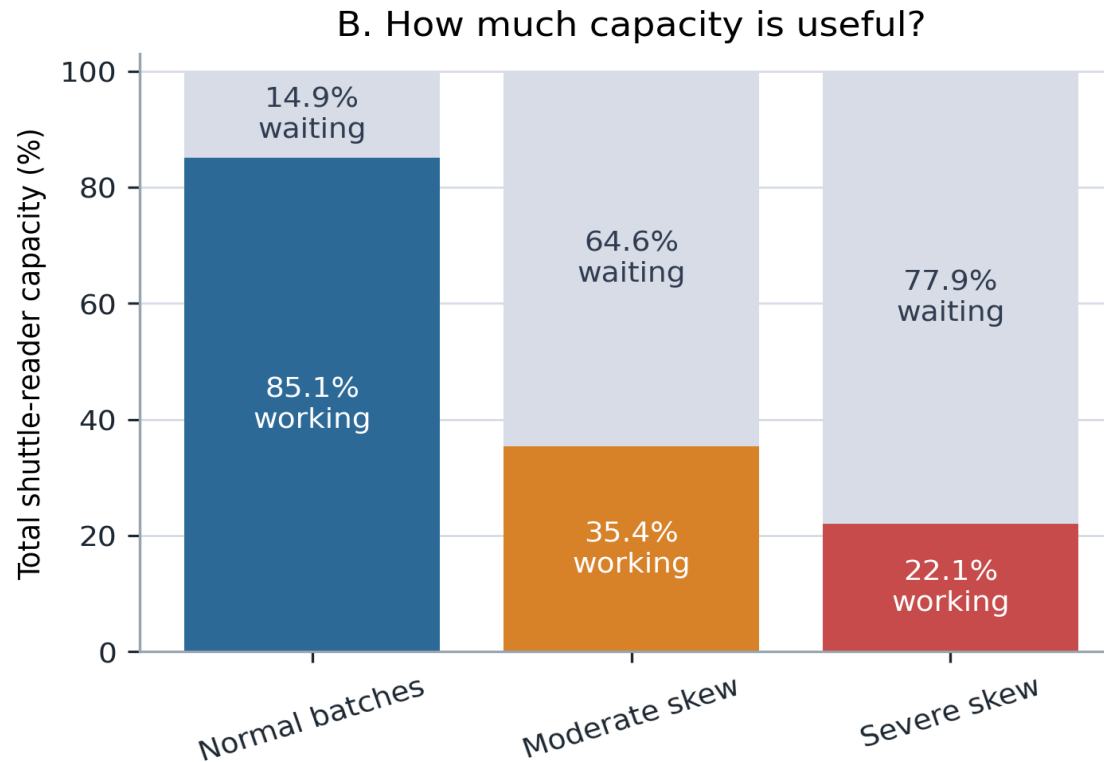
The experiment controls skew to study its impact. It does not claim that these exact skew levels naturally occur in every workload.

Logical Skew Is Not Physical Work Skew



- More logical requests can merge into fewer physical glass services.
- At the P95 batch, an 80% logical hot-zone share becomes **64.1% post-merge active work**.

Under Fixed Ownership, Skew Strands Capacity



- Useful shuttle-reader capacity falls from **85.1%** to **35.4%** and **22.1%**.
- Cold zones finish early while the hot-zone owner remains on the critical path.

The Performance Consequence

Workload	Makespan	Throughput	P99 latency
Normal order	15.29 h	1.817 req/s	0.79 h
Moderate skew	32.50 h	0.855 req/s	1.75 h
Severe skew	42.79 h	0.649 req/s	2.39 h

- Severe controlled skew increases makespan by **2.80×**.
- The bottleneck is **where active work is owned**, not a lack of aggregate resources.

Evidence Boundary: Batch Size Matters

- The 50% and 80% skew levels are **experimentally controlled**.
- The observed impact depends on this dataset, its locality, and merge behavior.
- A larger batch may combine more independent requests and **smooth short-lived skew**.
- However, this is not a universal monotonic rule:
 - persistent spatial hotspots may remain skewed;
 - merging can change physical work differently from logical request counts;
 - online arrivals may overlap instead of waiting for a batch barrier.

Batch size must be swept explicitly before claiming how often real workloads expose this problem.

Why Microsoft Uses Static Partitions

- The panel is divided into rectangular logical partitions.
- Each partition contains one shuttle and at least one read-drive slot.
- During normal operation, a shuttle remains inside its partition.

This is a deliberate engineering tradeoff:

- shorter and more predictable paths;
- simpler real-time traffic management;
- lower congestion exposure at read drives and partition boundaries.

Where Fixed Ownership Becomes Restrictive

Static partitions assign service responsibility according to **panel geometry**, while demand follows a **time-varying access pattern**.

When active requests concentrate in a few physical regions:

- hot-zone shuttles and readers remain busy;
- cold-zone resources become idle;
- completion is determined by the busiest owner;
- fixed service regions cannot follow a moving or bursty hotspot.

The mismatch is not partitioning itself. It is long-lived, geometry-driven ownership that cannot adapt to the active working set.

Work Stealing Restores Some Flexibility

Project Silica uses work stealing as a load-imbalance fallback:

- the controller monitors data volume across partitions;
- a lightly loaded shuttle may temporarily enter an overloaded partition;
- the shuttle uses shortest-path routing outside its normal region.

In Microsoft's skew evaluation:

- tail completion time improves from **over 21 h to 11.5 h**;
- tail travel time increases from **29.4 s to 76 s**.

The Remaining Work-Stealing Cost



- A lightly loaded helper may be far from the hot zone.
- Assistance includes repositioning, platter transport, and return travel.
- Intermediate zones and hot-zone ingress may experience additional traffic.

Key Insight: Load Alone Is Not Enough

In our collision-free single-helper experiment:

Helper	Hotspot makespan reduction	Extra aggregate travel
Nearby Zone 2	40.6%	0.68 h
Distant Zone 7	27.4%	1.91 h

Net assistance value
= completion-time reduction
– cross-zone movement
– interference and congestion risk

The current model measures direct movement cost; route-conflict waiting and multi-helper congestion remain future work.

Proposed Direction: Adaptive Service Zones

The goal is **not unrestricted per-request shuttle movement**.

Instead, adapt service responsibility to the active workload:

- resize or reshape service regions at each control epoch;
- assign more shuttle-reader capacity to active hot regions;
- avoid reserving full service capacity for inactive cold regions;
- keep each region spatially compact;
- bound cross-zone distance and concurrent helpers;
- use hysteresis to avoid frequent reconfiguration.

Research Hypothesis

Request-adaptive service zones can recover stranded capacity with less cross-region travel than reactive work stealing.

Required comparison on identical merged glass tasks:

1. Strict static partitions
2. Microsoft-style work stealing
3. Adaptive service zones
4. Ideal shared-pool upper bound

Primary metrics:

- makespan, throughput, and P95/P99 latency;
- shuttle-reader utilization balance;
- cross-zone travel and conflict waiting;
- reconfiguration cost and stability.

Takeaway

1. Controlled batch skew shows how fixed ownership can strand capacity.
2. The result is batch- and dataset-dependent, so sensitivity analysis is required.
3. Static partitions provide valuable locality and congestion control.
4. Work stealing restores flexibility, but adds reactive cross-zone movement.
5. The opportunity is to make **service regions adapt to active requests**, while keeping movement structured and bounded.

Adapt ownership to the workload, rather than forcing every workload into the same fixed ownership map.